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A Data-Driven Model for Predicting Competitive Adsorption of CH₄ and CO₂ in Shallow Coal Seams During CO₂-Enhanced Coalbed Methane Recovery

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Abstract

CO₂-Enhanced Coalbed Methane (CO₂-ECBM) has gained a lot of attention in recent times. Efficient methane recovery from Coalbed Methane (CBM) reservoirs depend on understanding the complex interactions of CH₄ and CO₂. During CO₂-ECBM, injected CO₂ adsorbs onto the coal surface and displaces CH₄ which enhances the methane production. This work proposes a novel machine learning (ML) tool for predicting the amount of adsorbed CH₄ and CO₂ during CO₂-ECBM process at various conditions.

An experimental dataset was gathered from the literature with various operational conditions, such as pressure, injected gas composition, and coal properties. Dataset pre-processing and analytics were initially conducted to ensure optimal inputs for ML modeling. Several ML tools, including Random Forest (RF), eXtreme Gradient Boosting (XGBoost), and Decision Tree (DT), were employed to predict gas adsorption. The hyperparameters were optimized using K-fold cross-validation, integrated with the grid search approach. The performance of ML models was evaluated using the following error metrics: coefficient of determination (R²), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Akaike Information Criterion (AIC), and Bayesian Information Criterion (BIC).

The total dataset was divided into an 80:20 ratio for model training and testing. Results show that XGBoost outperforms other models, showing R² more than 90%. The proposed model can predict gas adsorption (0 – 905.36 scf/ton) as a function of pressure (0 – 1507.33 psi), temperature (39.9 °C – 44.3 °C), ash content (19.94 – 41.17 %), fuel ratio (2.65 – 3.88), vitrinite (11.4 – 51.28 vol. %), liptinite (0 – 52.2 vol. %), inertinite (1.36 – 76.8 vol. %), CO₂ concentration (0 – 100 %) and gas type (1 = CH₄ and 2 = CO₂). The trained model was tested using a blind dataset to confirm its robustness. In addition, the proposed model was also compared to another set of unseen experimental datasets. The model reflects that CO₂ has a stronger adsorption affinity toward coal than CH₄, which may help enhance the methane recovery from CBM reservoirs. This approach enhances our understanding of complex gas-coal interactions in CO₂-ECBM processes.

This study presents a novel model for analyzing the competitive adsorption of CH₄ and CO₂ without the need for labor-intensive simulations or laboratory experiments. The proposed XGBoost model accurately predicts gas adsorption in CBM reservoirs, significantly reducing computation time. The findings highlight

the potential use of ML to optimize CO₂-ECBM strategies by improving methane recovery and reducing environmental impact by effective carbon sequestration.

Introduction

Reductions in carbon dioxide (CO₂) emissions are necessary to keep global warming below globally agreed targets (Intergovernmental Panel on Climate Change [IPCC], 2021). One strategy for mitigating climate change is carbon geological storage (CGS), which involves permanently storing CO₂ from major point sources. Various geological formations are being investigated as potential repositories for storage of captured CO₂, including (i) deep saline aquifers, (ii) depleted oil and gas fields, and (iii) unmineable coal seams. Storing CO₂ in geological formations not only contributes to achieving net-zero emissions, but in some cases, it also offers additional benefits. In particular, CO₂ injection into coal beds presents a dual advantage: it sequesters captured CO₂, stores it in coal, and in the process, enhances CH₄ production. Thus, CO₂-ECBM is the unique technology that attains both the ends of climate change mitigation and energy recovery, thereby representing a mutually beneficial approach to sustainable development (Zhu et al., 2025).

Coal seams are viable sites for CO₂ storage primarily due to their strong sorption affinity for it. CH₄ in the subsurface of a CBM reservoir exists in an adsorbed form on the inner surfaces of the coal framework. In the event of CO₂ injection, CO₂ competes for adsorption points and typically prevails due to its greater adsorption capacity and preference on the coal surfaces. The coal ends up selectively adsorbing the injected CO₂ and desorbs, or releases, adsorbed CH₄. The CH₄ is desorbed and produced, while CO₂ is stored primarily through adsorption onto the microporous surfaces of the coal matrix, thus mitigating both greenhouse gas emissions and enhancing gas recovery. The competitive adsorption process by which CO₂ preferentially adheres to coal (tending to adsorb in greater amounts than an equivalent amount of CH₄) is the principle on which CO₂ injection is applied to increase CBM production. Knowledge of and prediction of the interaction of CO₂ and CH₄ adsorption in coal are thus essential for the design of effective CO₂-ECBM applications. Pressure, temperature, and the characteristics of coal are among the important factors that affect the adsorption equilibria and kinetics for the two gases, and determination of conditions for optimal injection will thus require effective predictive models.

Traditionally, extensive numerical calculations and laboratory measurements were conducted to examine and predict the behavior of the CO₂-ECBM process. Coupled-reservoir simulations completely entrained the subtle physics of gas adsorption, diffusion, and flow in coal beds, yet they are accompanied by important downsides: first-principles simulation is often data-intensive, time-consuming, and computationally costly. High-fidelity models entail large numbers of parameters (e.g., multi-component isotherms, swelling effects, permeability evolution) and fine spatial and temporal discretization, with the outcome being extensive run time, particularly when sensitivity studies or large systems are involved (Alanazi et al., 2023). Similarly, in the laboratory, the determination of gas adsorption isotherms and other reservoir characteristics, which are necessary for calibration, is extremely labor-intensive and time-consuming (Zhang et al., 2020). Traditional methods are generally inflexible; a model calibrated to one formation or under a particular set of conditions will require important re-adjustment for use elsewhere. In operation, this implies that a new carbon sequestration candidate coal reservoir or sensitivity analyses (e.g., sensitivity to pressure/temperature conditions) using traditional simulations is extremely resource- and time-intensive. There is an obvious need for more effective, portable predictive tools to supplement or provide an efficient alternative to detailed numerical simulations and extensive experimentation in this area.

Data-driven modeling presents a potential solution to overcome the aforementioned limitations. Over the past decade, ML methods have become popular in petroleum and environmental engineering as powerful predictive tools because of their capacity to learn from existing data and capture non-linear interdependencies in it. Ensemble trees, including random forests and gradient boosting decision trees (with implementations including XGBoost), have set a new benchmark for modeling subsurface processes

in terms of accuracy and robustness. These approaches can also serve as rapid "surrogate" models that produce near-instant predictions upon training, and they are easier to generalize to different datasets than customized numerical simulators (Feng et al., 2021). To give a specific example, for modeling methane adsorption in coal, gradient boosting models have successfully simulated adsorption isotherms as a function of coal characteristics and replicated adsorption behavior in different domains with greater precision and generalization capacity than traditional models (Feng et al., 2021). These models effectively learn the predominant determinants of gas sorption (e.g., rank and composition of the coal, pressure) and can predict adsorption behavior under new conditions without having to loop through extensive simulations or experiments. By and large, the ML models are highly flexible: they can be updated and retrained as new information becomes available; therefore, they can be adjusted relatively unproblematically to different reservoirs or operational conditions.

Promising results have been reported from earlier applications of machine learning to related issues of coaled methane and CO₂ storage. For example, researchers have applied neural network and deep learning models to predict the CBM production rate and optimize the enhanced recovery processes. Xue et al. (2024) designed a hybrid data-driven model for CO₂ injection into coal seams that was able to predict important performance metrics (CH₄ production and CO₂ sequestration efficiency) and thus optimize the injection parameters accordingly. Likewise, Du et al. (2023) applied bidirectional long short-term memory (Bi-LSTM) to capture dynamic production patterns in CBM wells, with accurate production prediction. In the area of gas adsorption and storage, data-driven approaches have demonstrated success in approximating and projecting laboratory results. Zhang et al. (2020) applied various ML models to predict methane adsorption isotherms based on coal properties, with excellent consistency with laboratory measurements. A broad study by Feng et al. (2021) conducted comparative study of seven ML algorithms, including RF, Gradient Boost (GB), and XGBoost for the prediction of coal gas adsorption, concluding that ensemble learning methods produced robust and highly generalizable performance and suggesting the application of these methods for the rapid estimation of the adsorption capacity of methane.

Considering these advancements, in this research, we develop a data-driven framework for modeling competitive adsorption behavior of CH₄ and CO₂ in CO₂-ECBM processes. To build predictive models of the multicomponent gas adsorption in coal, we employed ensemble machine learning algorithms - specifically, XGBoost, DT and RF algorithms. These models were applied in an identification and selection process for suitable descriptors for adsorption capacity, subject to pressure, temperature, and coal properties, for both CO₂ and CH₄ in competitive conditions. We describe the complete modeling workflow, including data analysis, data preprocessing and hyperparameters tuning for the XGBoost, DT, and RF models. In addition, the performance of the constructed machine learning models was evaluated using error metrics and benchmarked against experimental data.

Methodology

Data description and analytics

The dataset used in this study was collected from literature, consisting of experimental studies (Asif et al., 2019, 2024). The employed coal samples belong to Jharia Coalfield with the coal rank ranges from sub-bituminous coal to low-to medium volatile bituminous coal. Complete details about the employed coal types and their nature are presented in Table 1.

Table 1—Data description.

Coal	Rank	Ash Content, wt. %	Fuel ratio	Vitrinite, vol. %	Liptinite, vol. %	Inertinite, vol. %
Coal A	Medium volatile bituminous	19.94	2.72	29.4	0	64.2
Coal B	Low volatile bituminous	41.17	3	11.4	0.2	76.8
Coal C	Medium volatile bituminous	36.25	3.08	20.8	0	73
Coal D	Medium volatile bituminous	24.92	2.81	16	4.6	52.6
Coal E	Low volatile sub-bituminous	30.41	2.65	40.33	52.2	1.36
Coal F	Medium volatile sub-bituminous	30.21	3.875	51.28	37.6	3.85

The experimental data was organized into a structured database with 272 data points. To develop and validate the model, the database was divided randomly into training and testing sets at a ratio of 80:20. A statistical overview of the training set is given in [Table 2](#), which shows the major descriptors - minimum, maximum, mean, standard deviation, and 25th, 50th, and 75th percentiles. The input parameters considered in this study include pressure, temperature, ash content, fuel ratio, volumetric proportions of vitrinite, liptinite, and inertinite, CO₂ composition, and gas type.

Table 2—Statistical analysis of the training dataset.

Parameter	count	mean	standard deviation	minimum	25 percent	50 percent	75 percent	maximum
Pressure, psi	217	616.11	450.03	0	248.04	511.79	1001.74	1507.33
Temperature, °C	217	42.92	1.65	39.9	41.69	44	44.18	44.3
Ash content, wt.%	217	30.09	6.32	19.94	24.92	30.21	36.25	41.17
Fuel Ratio	217	3.05	0.46	2.65	2.72	2.81	3.08	3.88
Vitrinite, vol. %	217	31.62	14.23	11.4	16	28.1	40.33	51.28
Liptinite, vol. %	217	20.98	22.48	0	4.6	4.6	37.6	52.2
Inertinite, vol. %	217	37.55	32.52	1.36	3.85	52.6	73	76.8
CO ₂ concentration, mol%	217	44.29	29.65	0	30	40	60	100
Gas Type	217	1.46	0.5	1	1	1	2	2
Gas adsorption (scf/ton)	217	157.36	143.89	0	49.07	141.24	218.43	905.36

Building upon the statistical overview, we next explored the relationships among the input variables and gas adsorption capacity using a correlation heatmap ([Fig. 1](#)). Of the features, pressure is moderately positively correlated ($r = 0.380$) with gas adsorption, indicating an increase in adsorption with increasing

pressure, consistent with the expected adsorption behavior. Conversely, the temperature is weakly positively correlated ($r = 0.069$), indicating a relatively minor effect on the range employed. Parameters describing the coal composition, such as fuel ratio and ash content, are moderately correlated with gas adsorption ($r = 0.243$ and $r = 0.247$, respectively), indicating the role played by coal maturity and quality in determining gas storage capacity. Of the petrographic attributes, inertinite is moderately positively correlated ($r = 0.259$) with gas adsorption, while weaker correlation values ($r = 0.137$ and $r = -0.196$) are found for vitrinite and liptinite, respectively. The large negative correlation ($r = -0.969$) amongst the percentages of vitrinite and liptinite in the maceral composition, i.e., the organic fractions — vitrinite, liptinite, and inertinite — of coal, emphasizes their anticipated inverse relationship. Additionally, concentration of CO_2 and gas type were strongly positively correlated ($r \approx 0.386$) with gas adsorption, indicating the stronger affinity of CO_2 over CH_4 toward the surfaces of coals. Overall, these results underscore the roles played by pressure, CO_2 concentration, and maceral composition in controlling the gas adsorption behavior in the studied coal samples.

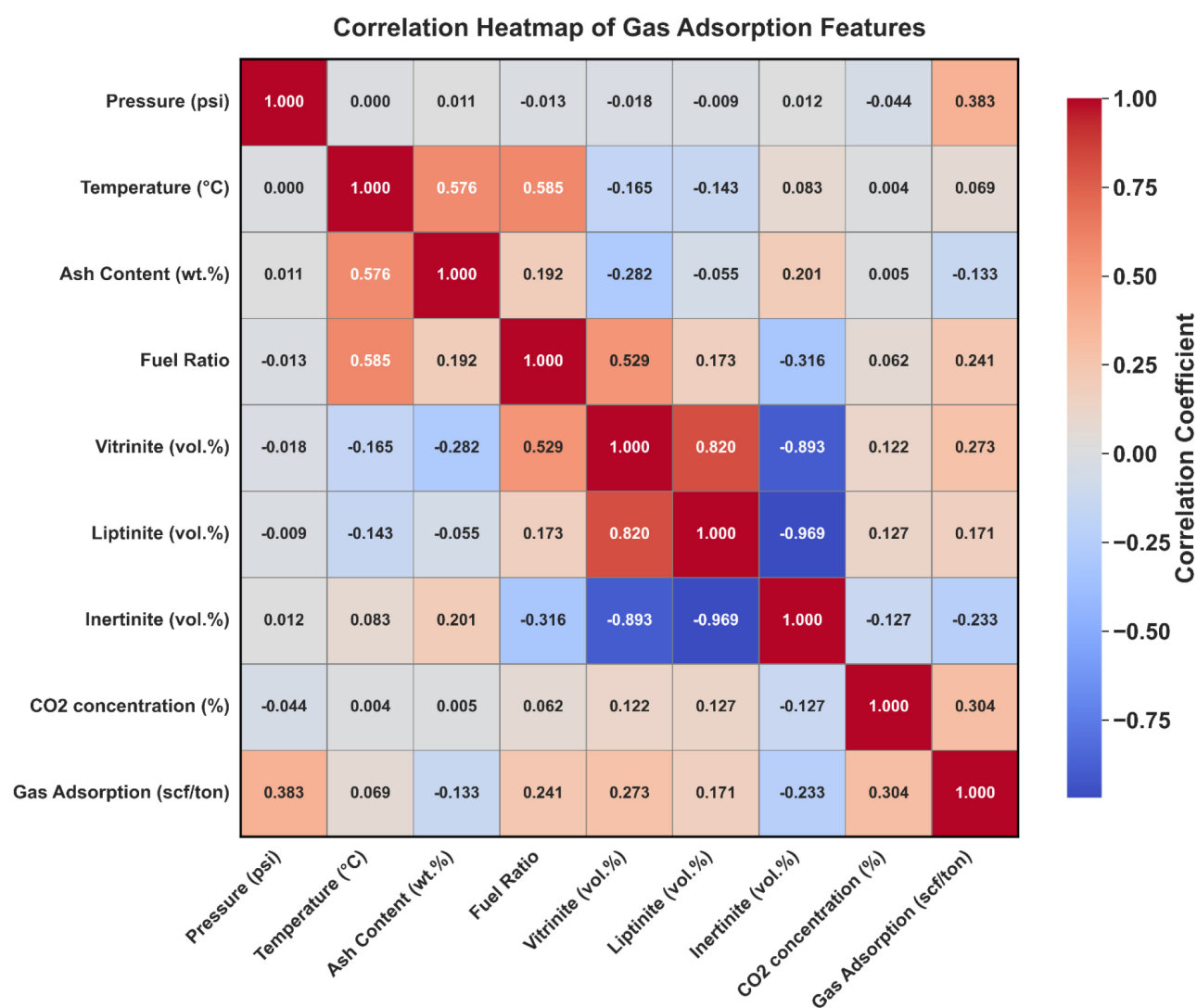


Figure 1—Correlation coefficient heatmap for the training dataset.

In addition to the correlation heatmap, a pair plot (Fig. 2) was generated to further visualize the relationships among features and their association with gas adsorption. The pair plot demonstrates how each feature in the dataset is related to gas adsorption. There is a lot of spread in pressure and CO_2 content,

and features such as ash content and maceral components vary less. The plots validate the observation that more gas adsorption is obtained with increasing pressure and that CO_2 is distinct from CH_4 in adsorption. Most of the other features are not significantly related to each other in any significant way, and no feature can effectively classify between CH_4 and CO_2 adsorption. This implies that multiple features should be used collectively in order to make accurate predictions. From the whole dataset, pressure, CO_2 content, and maceral composition were found to be the key factors influencing gas adsorption in the samples.

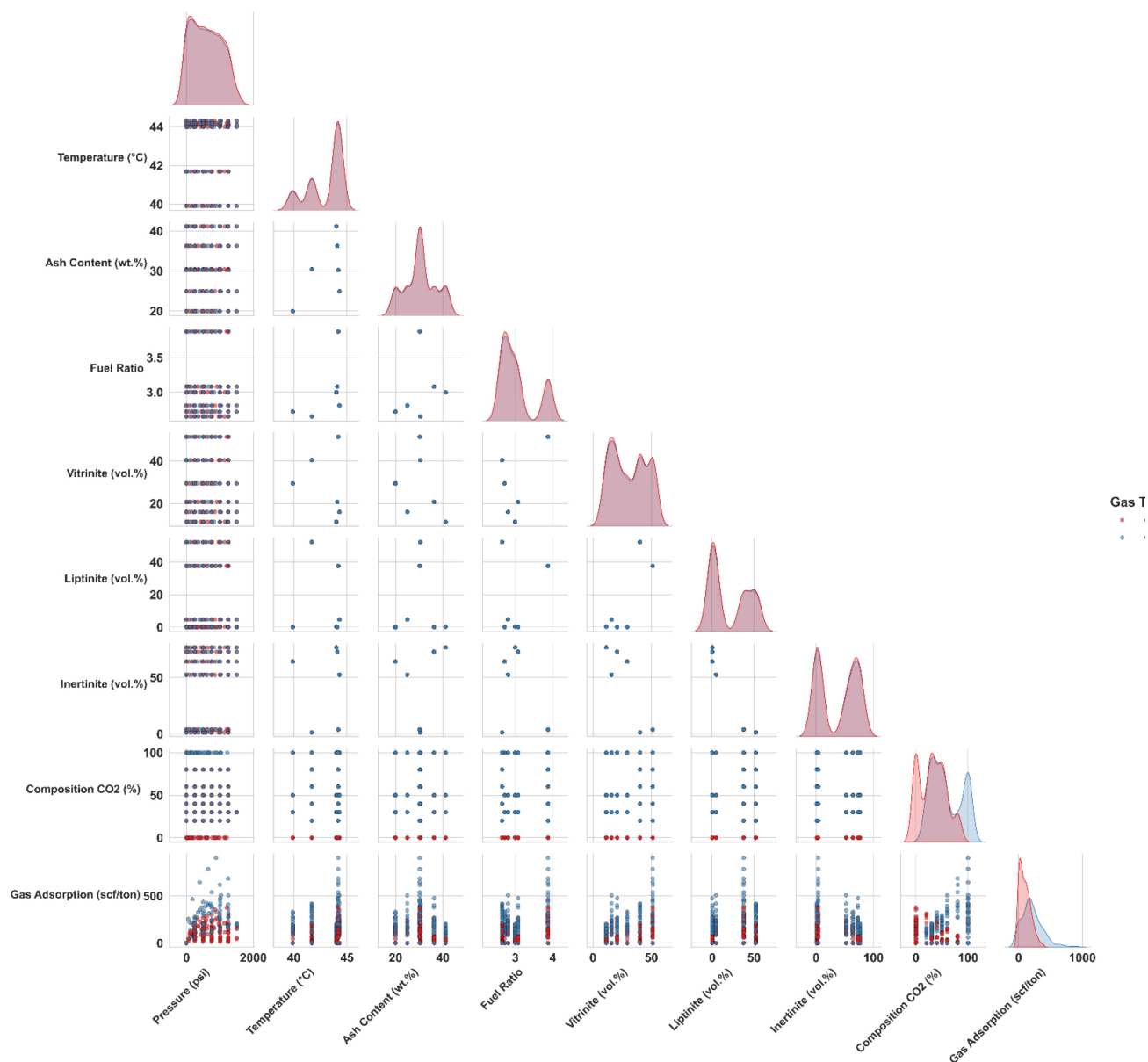


Figure 2—Pair Plot of gas adsorption features.

Error metrics

This study employed multiple machine learning models, XGBoost, DT and RF, to forecast coal gas adsorption. All models have different structures, and their prediction performance is influenced by this structure. The model performances were evaluated using a blend of error metrics: RMSE (eq. 1), R^2 (eq. 2), AIC (eq. 3), and BIC (eq. 4). Although RMSE and R^2 give simple indicators of model accuracy, they are not adjusted for the built-in uncertainties in the predictions. To move past this concern, AIC and BIC were utilized since they consider model likelihood and penalize model complexity, providing a stronger

assessment. AIC, with a 2-term penalty, is comparatively lenient with complex models, while BIC, with a sample size natural logarithm-based penalty, is a stricter complexity penalized (Ai, 1974; Bozdogan, 1987). Lower AIC and BIC values reflect good model fit.

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (X_{Ai} - X_{Pi})^2}{n}} \quad (1)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (X_{Ai} - X_{Pi})^2}{\sum_{i=1}^n (X_{Ai} - \bar{X}_{Ai})^2} \quad (2)$$

$$AIC = -2\ln(\text{likelihood}) + 2P_N \quad (3)$$

$$BIC = -2\ln(\text{likelihood}) + [\ln(n)]P_N \quad (4)$$

where: X_{Ai} — actual value, X_{Pi} — predicted value, $\bar{X}_{Ai}$ — average actual value, n — number of data points, P_N — number of parameters in the model.

Machine Learning Models

Random Forest. Random Forest algorithms are tree-based learning algorithms commonly used in classification and regression problems. In regression problems, the RF is an ensemble learning algorithm that learns by growing an ensemble of regression trees and training each tree on a bootstrap replica of the training data and predictors (referred to as input variables). RF employs bootstrap sampling and random feature selection to generate multiple distinct sets of decision trees, resulting in enhanced accuracy and robustness of the output prediction (Liaw & Wiener, 2002). More specifically, each tree in the forest was fitted to bootstrap samples (referred to as sampling bags) drawn from the training data. The nodes in each decision tree were split based on binary splits of randomly selected subsets of predictors, allowing the model to learn about intricate interactions among variables. The out-of-bag (OOB) samples not selected for the bootstrap were not used in the training of the particular tree. The OOB data were used as a form of natural validation set to assess the prediction error independent of any holdout data set. The OOB error was approximated by averaging the predictions from all trees for which some sample was not part of the training bootstrap. This built-in cross-validation is perhaps the greatest strength of the random forest algorithm because it can provide an unbiased estimate of the overall model performance (Breiman, 2001; Liaw & Wiener, 2002).

$$MSE_{OOB} = \frac{1}{N_T} \sum_{i=1}^{N_T} (x_i - \hat{x}_i)^2 \quad (5)$$

where: N_T is the number of OOB samples, x_i is the true value, and $\hat{x}_i$ is the OOB prediction for sample i .

Extreme Gradient Boosting. XGBoost is an ensemble learning algorithm based on decision trees that employs the gradient boosting algorithm in order to provide better prediction results (Chen & Guestrin, 2016). It uses an ensemble of Classification and Regression Trees (CART) to perform both regression and classification tasks. In contrast to a singular decision tree, which is prone to overfitting, new trees are added by the algorithm to the model in sequence, with each of the new trees learning from the existing ensemble to directly address the prediction errors (residuals) of the existing ensemble. The boosting is regulated by a regularized objective function that involves both a differentiable loss term (measuring prediction versus target difference) and a penalty term that regulates the complexity of the model to avoid overfitting. Mathematically, the objective to be optimized in the training at iteration s is represented as follows:

$$K^{(s)} = \sum_{i=0}^n m(y_i - \hat{y}_i) + \sum_{p=1}^P \beta(f_p) \quad (6)$$

where: y_i denotes the true target value for the i -th sample $\hat{y}_i$ is the prediction for that sample, f_p represents the function learned by the p -th decision tree in the ensemble, and $\beta(f_p)$ is a regularization term penalizing the complexity of tree f_p .

This regularized boosting framework allows XGBoost to achieve high accuracy while maintaining generalization by controlling model complexity (Chen & Guestrin, 2016).

Decision Tree. Decision trees are nonparametric supervised learning models employed in both regression and classification problems (L. Breiman; J. H. Friedman, 1984). A decision tree is a hierarchical tree in which decision tests on input features are performed by internal nodes, partitioning the data into progressively more homogeneous subsets recursively, and leaf nodes return final predictions (class labels or continuous values). The tree is built by employing the greedy, top-down recursive partitioning algorithm, in which, at each node, the feature and split threshold are selected to optimize some splitting criterion maximizing reduction of impurity or prediction error in the resulting subsets. In regression problems, splits are typically chosen to minimize the total squared residuals in each partition. Mathematically, in the case of the candidate split partitioning the samples into subsets S_1 and S_2 , the splitting criterion optimizes as follows:

$$\sum_{i \in S_1} (y_i - \bar{y}_{S_1})^2 + \sum_{i \in S_2} (y_i - \bar{y}_{S_2})^2 \quad (7)$$

where: y_i is the observed value of the i -th sample, and $\bar{y}_{S_j}$ is the mean target value in subset S_j .

Tree growth continues until criteria such as maximum depth or minimum leaf size are met, producing a model that partitions the feature space into regions associated with specific predictions (L. Breiman; J. H. Friedman, 1984).

Results and Discussion

Hyperparameter Tuning

Hyperparameter tuning was performed for each regression model (XGBoost, Random Forest, and Decision Tree) using GridSearchCV with 5-fold cross-validation. GridSearchCV employs an exhaustive search strategy that systematically evaluates all specified hyperparameter combinations, training, and validating models for each combination using a K-fold scheme. This approach ensures a thorough exploration of the hyperparameter space of each algorithm. For XGBoost, typical hyperparameters tuned include the learning rate, number of boosting rounds (param_n_estimators), maximum tree depth and regularization parameters, such as the minimum child weight (min_child_weight). For random forest, the key hyperparameters include the number of trees (param_n_estimators), the maximum depth of each tree (maximum_depth), the minimum samples required to split a node (min_samples_split), the maximum features considered at each split (max_features). For a Decision Tree regressor, we likewise tune parameters controlling the tree's complexity – for example, the maximum depth (maximum_depth), the minimum samples for splits (param_min_samples_split) and maximum features (max_features). Using 5-fold cross-validation (CV = 5) in the grid search provides a robust estimate of performance for each hyperparameter set: the data are partitioned into five folds, and the model is trained and validated on multiple fold combinations, with results averaged across folds. This cross-validated search procedure mitigates overfitting to any single train–test split and aids in selecting a model configuration that generalizes well to unseen data. Consequently, the exhaustive tuning process identified the best-performing hyperparameter combinations (Table 3), which significantly improved the models' accuracy and generalization ability over non-optimized default settings.

Table 3—Hyperparameter statistics of the applied machine learning models

ML Tool	Hyperparameter	Values / Range	Optimized value
XGBoost	param_n_estimators	10-36	36
	maximum depth	1-9	9
	learning rate	0.1	0.1
	min child weight	1-9	3
Random Forest	param_n_estimators	100-1500	750
	max_features	{‘sqrt’, ‘auto’, ‘log2’}	‘sqrt’
	maximum depth	2-10	10
	min_samples_split	2-10	2
Decision Tree	max_features	{‘auto’, ‘sqrt’, ‘log2’}	‘sqrt’
	param_min_samples_split	2-10	5
	maximum depth	15-20	15

Table 4 presents the performance measures of the XGBoost, Random Forest, and Decision Tree models on both training and testing data. For each model, important evaluation metrics are provided: RMSE, R^2 , AIC, and BIC. On the training data, XGBoost had the lowest RMSE (15.56 mmol/g) and the highest R^2 (0.988), reflecting excellent fit and prediction quality, in addition to giving the lowest AIC and BIC values reflecting quality of the models and penalizing complexity. On testing data, not only is XGBoost better than the other models in having the lowest RMSE (40.70 mmol/g), the highest R^2 (0.927), and AIC and BIC values below those of Random Forest and Decision Tree, but these results clearly illustrate not only that XGBoost fits the training data well but also generalizes best to novel, unseen data. Thus, we conclude that XGBoost is the strongest and most accurate model among the models tested in this research.

Table 4—Error metrics for machine learning models.

		XGBoost	Random Forest	Decision Tree
Training data	RMSE, mmol/g	15.564	19.129	40.559
	R^2 , fraction	0.988	0.982	0.924
	AIC	1211.324	1300.816	1427.049
	BIC	1245.123	1334.615	1459.519
Testing data	RMSE, mmol/g	40.704	46.286	61.728
	R^2 , fraction	0.926	0.905	0.806
	AIC	427.695	441.832	696.130
	BIC	447.769	461.906	720.197

Fig. 3 presents the scatter plot of the actual vs predicted gas adsorption (scf/ton) of the training (a) and testing (b) data by the Random Forest, XGBoost, and decision tree models. The prediction of either the training or testing data is represented by each point, where different markers separate the models and data splits. The closeness of points toward the 45-degree line represents the goodness of predictions with better model performance, as shown by points near the line. We observed that the XGBoost and Random Forest models, in particular, have tighter clustering of both training and testing data points along the perfect prediction line, which represents better prediction and generalizability of the models. This is also

demonstrated by their AIC and BIC having lower values, reflecting better fitting and less complexity of the models. The results clearly illustrate how ensemble models such as XGBoost and Random Forest are more reliable and consistent in the prediction of gas adsorption than the individual decision tree model.

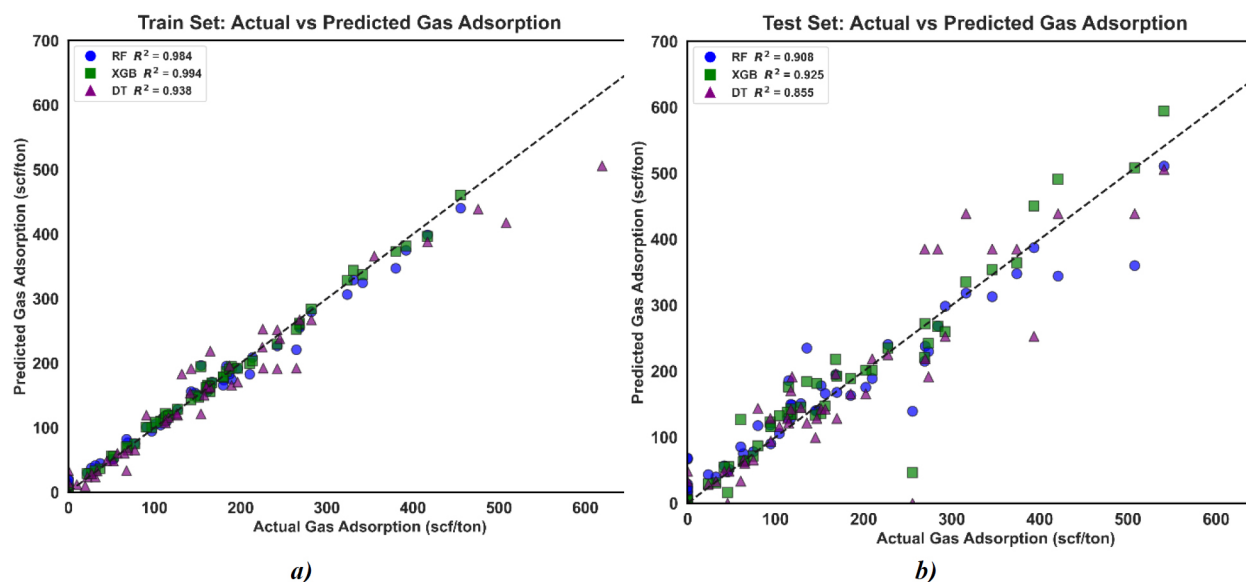


Figure 3—Predicted vs. Actual gas adsorption (scf/ton) of three ML Models on Training (a) and Testing Data (b).

Sensitivity analysis

Effects of coal type. Figs. 4 and 5 provide a comparison between experimental and XGBoost-predicted isotherms of gas adsorption on six different types of coal under pure methane and pure CO₂, respectively. In both graphs, adsorption increases with pressure in all cases, and the trend within each coal type is different, illustrating the differences between them. Most importantly, the prediction lines of the XGBoost model adhered very closely to the experimental data in all pressure ranges, affirming the strong prediction capability of the model and its replication of the true adsorption behavior. Coal F is only to display the greatest adsorption capacity toward both methane and CO₂, while Coals A and B retain the lowest. Coal F was found to have the greatest vitrinite content (51.28%) and fuel ratio (3.875), with moderate liptinite content (37.6%) and the lowest inertinite fraction (3.85%) (Table 1). These maceral constituents are found to play a crucial role in adsorption and are characterized by features such as elevated vitrinite and fuel ratio in the case of enhanced adsorption, whereas elevated inertinite content, found in Coals A, B, and C, tends to be responsible for lowered adsorption (Fig.4 and Fig.5). The differences in adsorption among the different types of coal can therefore be attributed to their maceral and ash contents. This emphasizes the pivotal role played by the properties of coal in the storage behavior of gas and the efficiency of the developed XGBoost model in identifying such intricate relationships and making accurate predictions regarding adsorption from coal properties.

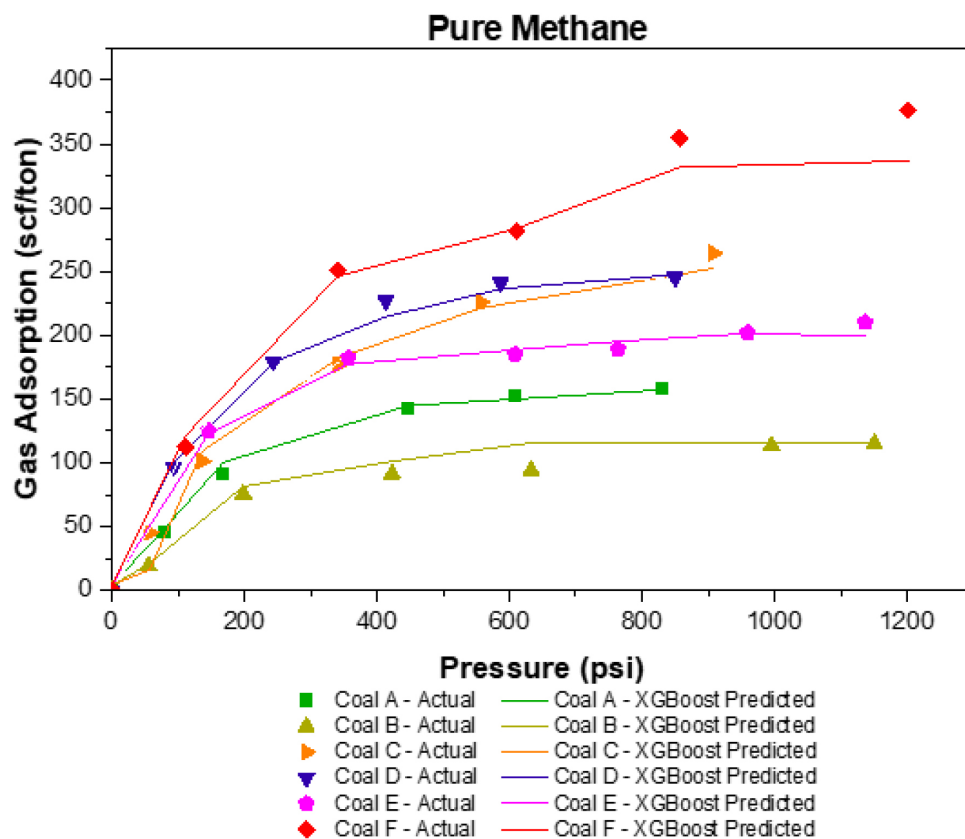
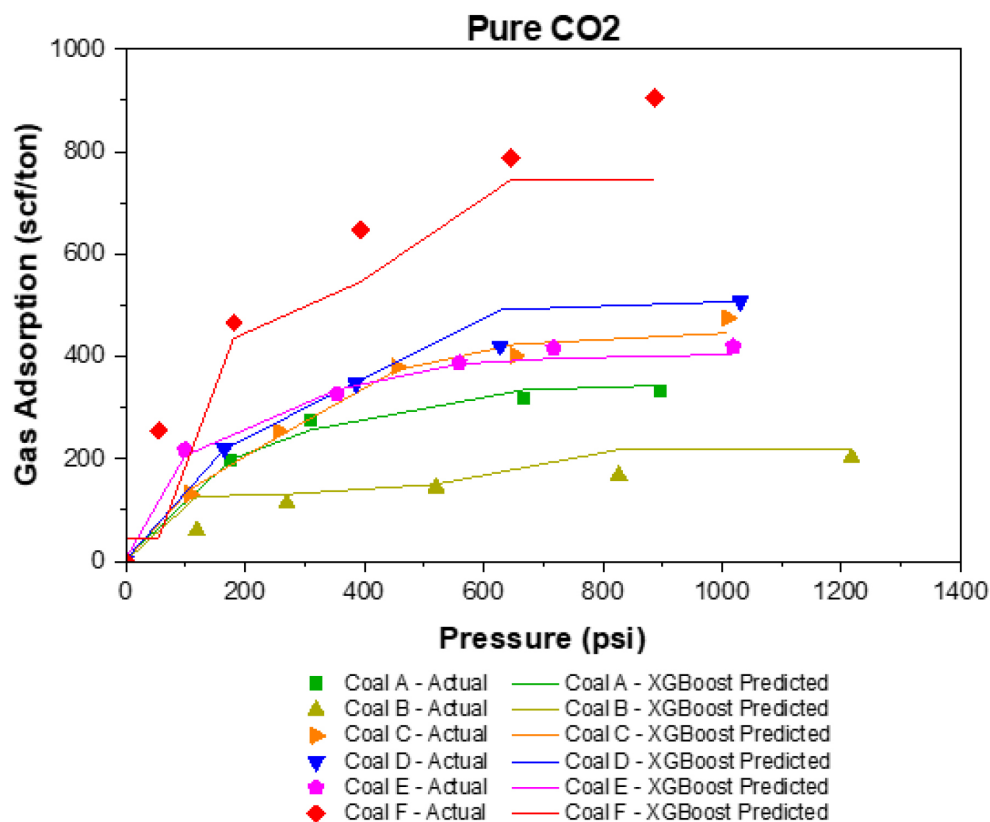


Figure 4—Pure Methane adsorption isotherms.

Figure 5—Pure CO₂ adsorption isotherms.

Effect of gas type. The gas adsorption characteristics of coal differ extensively with respect to both the gas type and inherent coal properties. Fig. 6 shows the experimental vs XGBoost-predicted adsorption isotherms for methane and CO₂ on four types of coals (A, B, C, and D) as a function of pressure. Each subplot is the adsorption capacity (scf/ton) versus pressure (psia), with the adsorption isotherms of methane in blue and those of CO₂ in red; closed symbols are actual measurements, and the open symbols are the predictions of XGBoost. In each instance, CO₂'s adsorption exceeds that of methane by large margins at all pressures, reflecting its greater coal affinity. The adsorption increased with pressure, particularly in the case of CO₂. Coals A and D clearly demonstrate the best match between XGBoost predictions and experimental data for both methane and CO₂, with the lines of prediction nearly perfectly overlapping with actual data for all pressure points. In the case of Coals B and C, the predictions by the model are also well in accordance with the experimental behavior, with minor deviations seen in the case of methane at higher pressure values. Overall, the close concordance of XGBoost predictions with observed data, particularly for coals A and D, reflects the strength of the model in identifying the adsorption tendencies of gases, and the differences in adsorption values among coals reinforce the key roles of gas type, pressure, and properties of the specific coal.

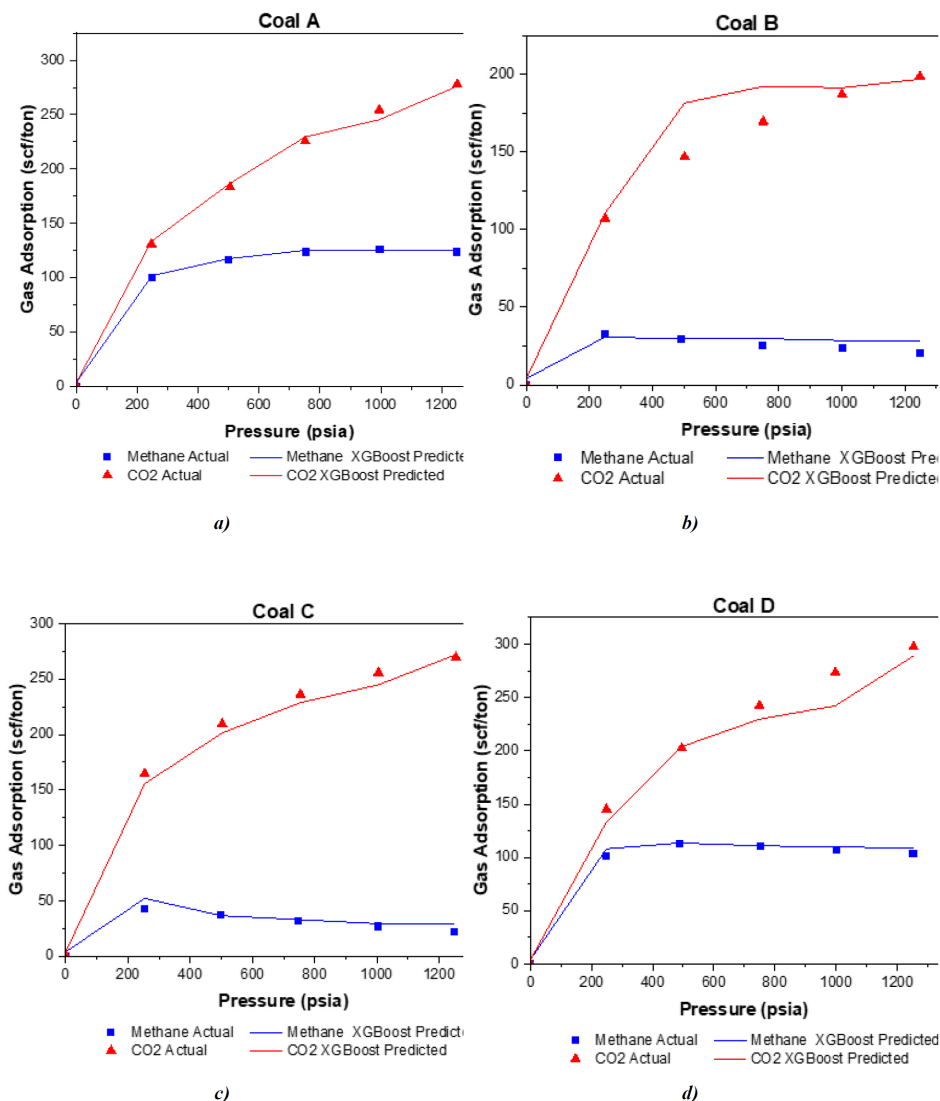


Figure 6—Adsorption equilibria in a binary system of methane and carbon dioxide (50% each).

Conclusions

This work developed robust data-driven predictive models for the competitive adsorption of CO₂ and CH₄ during the CO₂-ECBM process using advanced machine learning methods. An extensive experimental dataset with the pressure ranging from 0 to 1507 psi, gas composition ranging from 0 to 100% CO₂/CH₄, as well as coal property ranges (ash from 19.94 to 41.17 wt.%, fuel ratio from 2.65 to 3.88), was utilized. Nonetheless, the temperature range (39.9–44.3°C) is relatively narrow, while the subsequent work will expand the model to a wider temperature range to further enhance its applicability. Among the models tried, XGBoost had the optimum performance compared to random forest and decision tree models. Particularly, using the test dataset, the XGBoost model had an R² value of 0.926, an RMSE value of 40.7 scf/ton, and the minimum values of the AIC (427.7) and BIC (447.8) compared to all the models, which reflect good predictability and generalizability. On the training dataset, the model also had an R² value of 0.988 as well as an RMSE value of 15.6 scf/ton, with correspondingly small values of the AIC as well as the BIC.

These findings unequivocally show the critical role played by pressure, CO₂ concentration, and maceral composition (most importantly vitrinite and inertinite concentrations) in governing the adsorption characteristics. The XGBoost model effectively identified nuanced nonlinear relationships between various input features and accurately replicated the CH₄ as well as the CO₂ isotherms in different coals, even in unseen test sets. Sensitivity analysis verified the significant effect that variations in maceral composition have on gas adsorption capacity. Increasing vitrinite and decreasing inertinite content were related to higher adsorption capacity. These findings affirm the robust predictability inherent in the use of machine learning, especially ensemble tree-based methods, in the estimation of multicomponent gas adsorption in coal seams without the need to employ time-consuming experiments or computer-intensive simulations.

In all, the developed model offers an effective and accurate means to optimize CO₂-ECBM operations, enable methane production, and maximize CO₂ storage in unminable coal seams. Subsequent work will involve increasing the dataset with wider operational parameters, geology variety, and field data, which will strengthen the model, making it even more robust, pragmatic, and applicable in industrial applications. Such initiatives will have the capability to make significant impacts towards optimizing methane production as well as sustainable energy development.

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